

GeoSentinel AI — Analysis Report

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Analysis Period

Field	Value
Date 1 (T1)	2016-01-09
Date 2 (T2)	2026-07-09
T1 Scene ID	—
T2 Scene ID	—
T1 Acquisition	—
T2 Acquisition	—
T1 Cloud Cover	—
T2 Cloud Cover	—
Satellite	Sentinel-2 L2A
Resolution	10 metres
Processing Time	— s

Land Cover Area Change

Class	T1 Area (km²)	T2 Area (km²)	Change (km²)	Change (%)
Background	0.000	0.002	+0.002	+1600.0%
Urban	1.333	1.431	+0.098	+7.4%
Vegetation	1.790	1.767	-0.023	-1.3%
Water	13.173	13.020	-0.152	-1.2%
Barren	0.216	0.272	+0.056	+26.0%
Agriculture	0.089	0.108	+0.019	+21.2%

Recommendations

[CRITICAL] Urban Encroachment Near Water Bodies

Why: New urban pixels were detected within 0 m of water bodies (0.00 km² affected). Encroachment within the 500 m buffer is restricted under water body protection regulations.

Action: Urban expansion is occurring within 500 m of a water body. This is a high-risk scenario for flooding, contamination, and ecosystem degradation. Apply planning restrictions under Section 12 of the AP Water Bodies Protection Act. Initiate immediate demarcation of Full Tank Level (FTL) boundaries.

[HIGH] Rapid Urban Expansion Detected

Why: Urban cover increased by 9.1% (1.45 km²) between 2016-01-09 and 2026-07-09, exceeding the rapid-expansion threshold of 10%.

Action: Rapid urbanization is occurring in the study area. Engage HMDA (Hyderabad Metropolitan Development Authority) to review Master Plan compliance. Prioritize green buffer zones and ensure

infrastructure (roads, water, drainage) is upgraded proportionally.

[HIGH] Land Cover Change Hotspots Identified

Why: 49 change hotspot(s) were detected, each exceeding 0.5 km². The largest hotspot covers 0.12 km² centered at 17.3850°N, 78.4867°E.

Action: Concentrated change hotspots have been identified. These may indicate targeted development activity or environmental disturbance. Dispatch field teams to highest-priority locations for immediate ground verification and documentation.